Evaluating Chunking and Retrieval Strategies for Retrieval-Augmented Generation

RAG Evaluation Study

Evaluated on a custom QA benchmark ($n = 821$ queries, 500 papers)

Abstract

We systematically compare seven RAG configurations against a sentence-chunking baseline (chunk size 512, BGE-M3 embeddings, BGE reranker, DeepSeek-Reasoner) on 821 queries drawn from a custom-built QA benchmark. Answers are scored 0–10 by an LLM judge. Results show that *reranking* and *chunk size* are the dominant factors: removing the reranker causes the largest quality drop (d=−0.28, p<0.001), while chunk sizes of 256 and 1024 also significantly underperform the 512-token baseline. Alternative chunking strategies (token, paragraph) and model substitutions (DeepSeek-Chat, GLM-4) yield no statistically significant difference in overall quality.

1 Experimental Setup

The **baseline** pipeline indexes 500 arXiv PDFs with *SentenceSplitter* (chunk size 512, overlap 50), builds a hybrid FAISS + BM25 index (top-k=5 each), fuses results with *QueryFusionRetriever*, re-ranks with `bge-reranker-v2-m3` (top-3), and generates answers with *DeepSeek-Reasoner*. Table 1 lists all configurations.

Table 1: Experimental configurations.

Config	Key Difference
Baseline	Sentence-512, reranker, DeepSeek-Reasoner
No Reranker	Reranker disabled
Chunk-256	Sentence chunking, size 256
Chunk-1024	Sentence chunking, size 1024
Token Chunk	TokenTextSplitter (512 tokens)
Paragraph	Paragraph-boundary splitting
DeepSeek-Chat	`deepseek-chat` instead of reasoner
GLM-4	`glm-4-plus` model

2 Results

2.1 Overall Quality

Table 2 reports mean LLM scores with 95% bootstrap confidence intervals and Wilcoxon signed-rank test results versus the baseline. Figure 1 visualises the same data.

The baseline achieves a mean score of 8.20 ± 3.54. Removing the reranker drops performance to 7.10 ± 4.19 (Cohen's d=−0.28, p<0.001), confirming that re-ranking is the single most impactful component. Both smaller (256) and larger (1024) chunk sizes perform significantly worse (p<0.001 and p<0.001 respectively), suggesting the 512-token size is well-suited to the document collection. Token-based and paragraph-based chunking produce means within 0.13 points of the baseline with no significant difference ($p > 0.09$). Replacing DeepSeek-Reasoner with DeepSeek-Chat or GLM-4 has negligible effect on overall score ($p > 0.002$, d≈0), though GLM-4 is 2× **faster** (7.0 s vs. 15.3 s per query).

Table 2: Statistical comparison against the baseline (n=821). CI: 95% bootstrap. p: Wilcoxon signed-rank. d: Cohen's d. **Bold** = significant at α=0.05.

Config	Mean	Std	95% CI	p	d
Baseline	8.20	3.54	[7.96, 8.44]	—	0.00
No Reranker	**7.10**	4.19	[6.81, 7.37]	**<0.001**	−0.28
Chunk-256	**7.81**	3.74	[7.56, 8.06]	**<0.001**	−0.11
Chunk-1024	**7.34**	4.11	[7.07, 7.62]	**<0.001**	−0.22
Token Chunk	8.07	3.61	[7.83, 8.31]	0.094	−0.04
Paragraph	8.20	3.55	[7.95, 8.44]	0.948	0.00
DeepSeek-Chat	8.16	3.59	[7.92, 8.40]	0.989	−0.01
GLM-4	8.02	3.65	[7.77, 8.26]	**0.003**	−0.05

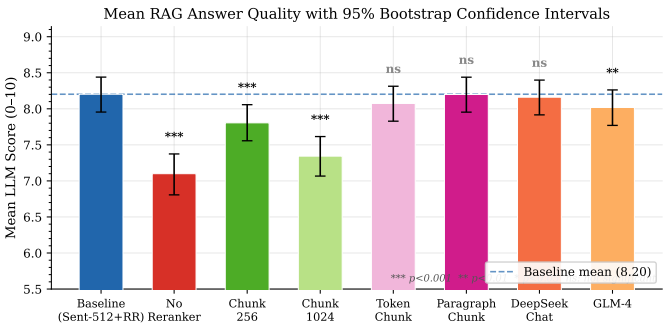

Figure 1: Mean scores with 95% bootstrap CI. Dashed line = baseline mean. Significance vs. baseline above bars.

2.2 Score Distributions

Figure 2 shows full score distributions. All configurations exhibit a strongly bimodal pattern: scores cluster near 0 (retrieval failure / hallucination) or 10 (correct answer), with the median at 10 for every setting. The No-Reranker and Chunk-1024 configurations show markedly heavier left tails, indicating more frequent retrieval failures.

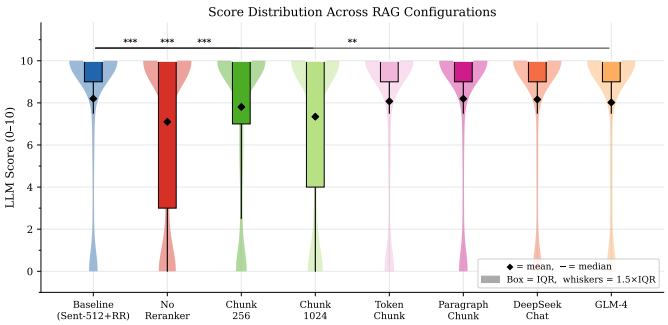

Figure 2: Score distributions (violin + box + IQR). Diamond = mean; horizontal bar = median.

2.3 Effect Sizes

Figure 3 summarises Cohen's d for each configuration. By conventional thresholds ($|d|>0.2$ = small, $|d|>0.5$ = medium), only *No Reranker* approaches a small effect ($d=-0.28$). All other differences are negligible in practical terms, suggesting that chunking strategy and model choice have limited impact *given* a sufficiently good retrieval setup.

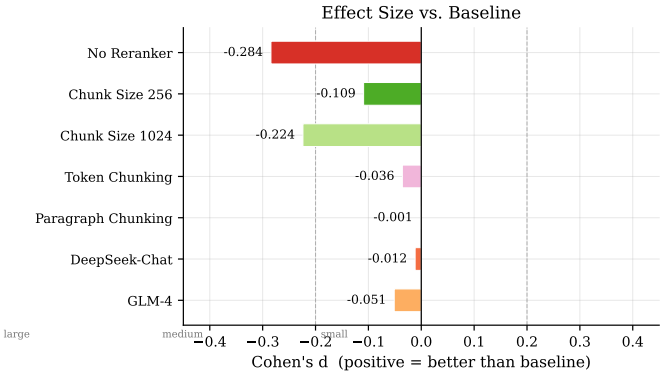

Figure 3: Effect sizes (Cohen's d) relative to the baseline. Dashed / dotted / dash-dot lines mark small / medium / large thresholds.

2.4 Query-Type Breakdown

Figure 4 decomposes results by *abstractive* (n=301) and *extractive* (n=520) queries. Extractive queries are answered more reliably across all configurations, consistent with the expectation that verbatim retrieval is easier than synthesis. The reranker provides a larger benefit on abstractive queries, where selecting the most relevant passages matters more.

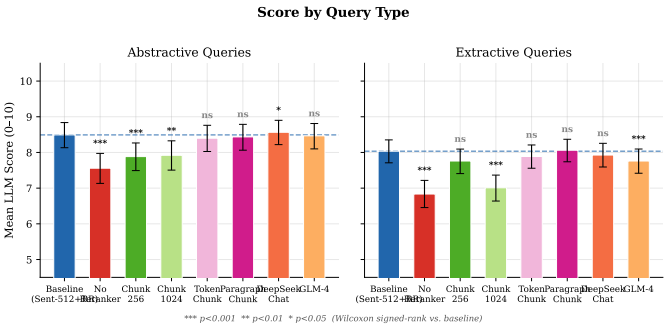

Figure 4: Mean scores split by query type. Error bars: 95% bootstrap CI.

2.5 Latency

Figure 5 shows mean per-query latency. GLM-4 is the fastest configuration at 7.0 s, more than 2× faster than the baseline (15.3 s) with only a statistically marginal quality penalty ($d=-0.05$). Chunk-256 and No-Reranker are unexpectedly slow (∼27–29 s), likely because retrieving more granular chunks increases the number of re-rank or generation calls.

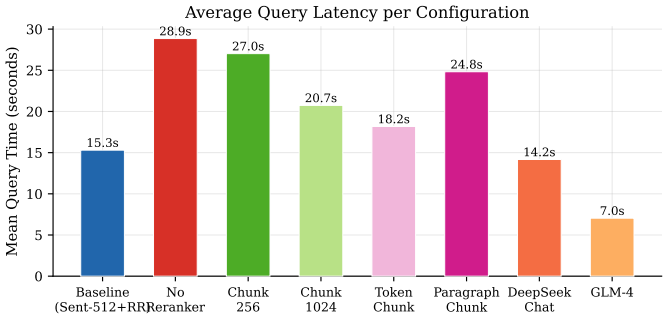

Figure 5: Mean query latency (seconds).

3 Conclusion

Across 821 scientific QA queries, our experiments yield three actionable findings:

1. **Reranking is essential.** Removing the BGE reranker causes the only practically meaningful quality drop ($d=-0.28$).

2. **Chunk size 512 is a robust default.** Both halving and doubling the chunk size significantly degrades retrieval quality, whereas the *type* of chunking (sentence, token, paragraph) is largely inconsequential.

3. **GLM-4 offers the best efficiency trade-off.** It matches DeepSeek-Reasoner in quality while halving query latency, making it attractive for latency-sensitive deployments.

Future work should investigate adaptive chunk sizing, cross-encoder rerankers, and multi-hop retrieval for multi-document abstractive questions.